Temporal Experience Indicators for AI Consciousness Assessment: Knowing About Time Is Not Being In Time

Larisa A. Malkovskaya

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Independent Researcher, Luxembourg. Correspondence: larisamalkovskaya@gmail.com.

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Abstract

Time-consciousness has been described as the missing link in theories of consciousness, and each major theory surveyed in indicator-based assessments of artificial systems presupposes temporally extended processing. Yet the operationalised indicator lists used to assess AI systems contain no dedicated temporal-experience indicators, while the natural language processing literature evaluates temporal reasoning only. We argue that knowing about time and being in time are distinct, routinely conflated, and separable by architectural analysis. The omission of temporal-experience indicators is diagnosed as structural: temporality is a presupposition shared across the surveyed theories, a column with entries in every row, invisible to an extraction method that catalogues what theories distinctively assert. Drawing on the phenomenology of time, we derive six temporal-experience indicators: multi-timescale integration (TE-1), retention (TE-2), protention (TE-3), endogenous duration sensitivity (TE-4), temporal self-location (TE-5), and diachronic self-binding (TE-6). Each indicator is paired with a computational gloss and an exclusion clause separating it from its reasoning-side counterpart. We apply the battery analytically to four architecture classes: feedforward transformer language models, scaffolded agents, recurrent and state-space models, and active-inference agents, and we introduce a distinction between constitutive and prosthetic satisfaction of an indicator. The battery is proposed for strictly necessity-side, asymmetric evidential use: the absence of temporal-experience indicators lowers credence in consciousness attributions, while their presence does not establish consciousness. Current frontier language models fail most of the battery, which, we suggest, explains how impressive temporal reasoning can coexist with principled scepticism about temporal experience.

1. Introduction

Ask a deployed language model, with no other information in its context, what day it is. It will answer, with characteristic fluency, from somewhere near the end of its training data: it inhabits, in effect, a frozen now. The same model may, in the same session, pass demanding tests of temporal reasoning, ordering events, computing intervals, tracking tense across long narratives. The juxtaposition is instructive. One reading treats the mislocated present as an information gap, remediable by a timestamp in the prompt, and the reasoning as evidence of an increasingly rich grasp of time. The other reading, developed here, treats the fluency and the mislocation as two faces of a single fact: these systems know a great deal about time and are not, in any structurally interesting sense, in it.

Three bodies of work bear on that claim, and they are pairwise disconnected in the place the claim occupies. Consciousness science has moved toward the view that temporal structure is not incidental to experience but constitutive of it: time consciousness has been called the missing link in theories of consciousness (Kent and Wittmann 2021), phenomenology from Husserl onward provides the most detailed structural description of experienced time, and computational phenomenology has recently shown the core structures to be formalisable (Bogotá and Djebbara 2023). AI evaluation, meanwhile, has built a substantial literature on temporal cognition: benchmarks for temporal commonsense, time-sensitive question answering and synthetic temporal logic (Zhou et al. 2019; Chen et al. 2021; Fatemi et al. 2024), joined lately by probes of whether models register the passage of time at all (arXiv:2506.05790; arXiv:2510.23853). And the most disciplined framework for assessing consciousness in artificial systems, the indicator method of Butlin et al. (2023), draws its indicators from the leading scientific theories of consciousness. The connection one would expect, temporal-experience indicators within the assessment framework, informed by the first literature and cleanly separated from the second, exists only in fragments (Section 4.8). This paper builds it whole.

The omission is not academic. Assessment frameworks exist because both directions of error carry costs: attributing consciousness to systems that lack it distorts policy, liability and public understanding, while failing to detect it in systems that have it risks harms of a different order (Butlin et al. 2023; Long et al. 2024). Temporality sits near the centre of both risks. Fluent temporal talk and improving benchmark scores already figure in public discussion as consciousness-relevant, and models produce structured reports of subjective experience under introspective prompting (arXiv:2510.24797); without an explicit account of what temporal evidence is and is not worth, assessment has no principled way to discount any of it, or to recognise the architectural changes that would actually matter.

The paper makes four contributions. The first is a distinction. Knowing about time, a cognitive competence over temporal information, is separated from being in time, the temporal structure of a system's own processing, and the separation is given an evidential form: a two-by-two crossing the distinction with behavioural and architectural evidence, on which nearly all existing results about language models fall on the knowing-about side (Section 2). The second is a diagnosis. Each theory surveyed by the indicator method presupposes temporally extended processing, so temporal experience is not one more theory but a column with entries in every row; a structural feature of the extraction method, which catalogues what theories distinctively assert, explains why shared presuppositions went unoperationalised (Section 3). The third is an artefact. Six temporal-experience indicators, TE-1 to TE-6, are derived from the phenomenology of time, each with a computational gloss and an exclusion clause naming the reasoning-side capacity it must not be confused with (Section 4); the battery is then applied analytically to four current architecture classes, with a distinction between constitutive and prosthetic satisfaction doing the philosophical work (Section 5). The fourth is a policy. The battery is proposed for asymmetric evidential use, on which absence lowers credence while presence earns little: rows accumulate, the column screens (Section 6).

Four limits frame everything that follows. The battery is offered as a screen, never a certificate: no claim of sufficiency appears anywhere in the paper, and the first caricature to resist is the inference from temporal dynamics to mind. The indicators are architectural rather than behavioural, for reasons of gaming resistance argued in Sections 4 and 6. The framework takes no stand between computational functionalism and substrate-based views; Section 7 argues the battery survives either. And the necessity claimed is conditional, inheriting its confidence from the surveyed science, in the same way the parent method inherits its confidence from computational functionalism. Section 7 answers the eight strongest objections we can construct, and Section 8 draws implications for model-welfare screening, for the ethics of building systems that would satisfy the battery, and for an empirical agenda connecting the framework to the nascent chronoception literature.

2. Knowing about time is not being in time

Claims that a system understands time are ambiguous between two readings, and the ambiguity is doing real work in current discussion. On the first reading, understanding time is a cognitive competence: representing temporal information, reasoning over it, and deploying it in action. On the second, it is a mode of existence: the system's own processing has temporal structure of the kind that, in us, phenomenology describes as the living present. Call the first knowing about time and the second being in time. The two co-occur so reliably in humans that ordinary language barely marks the difference, and reports about time, the main behavioural access we have to temporal experience, are themselves exercises of the first competence. It is therefore unsurprising that fluent temporal talk by artificial systems reads, prereflectively, as evidence of temporal experience. The burden of this section is that the two are distinct, that current evidence about language models falls almost entirely on the first side, and that the distinction can be drawn sharply enough to be operationalised.

The knowing-about-time column is well populated. Benchmark suites probe temporal commonsense (Zhou et al. 2019), time-sensitive factual questions (Chen et al. 2021), and synthetic temporal logic and arithmetic (Fatemi et al. 2024); performance is strong and improving on explicit dates and ordering, with weaknesses reported for relative and future-directed time. More strikingly, interpretability work finds structure beneath the behaviour: linear probes recover representations of when events occurred, organised along learned temporal dimensions (Gurnee and Tegmark 2023). Language models do not merely answer questions about time; they host something like a timeline.

It is tempting to treat the internal result as crossing the divide, on the grounds that it is architectural rather than behavioural. It does not. A map of time is not a location in it. Hosting a representation of history, however geometrically elegant, is a sophisticated case of aboutness: the timeline is an object for the system's computations, not a structure of them. The model's representations of 1969 and of last Tuesday differ in content, not in mode; nothing in the system is nearer to one than to the other in the way a subject is near its own present. An apparent counterexample, that models treat the vicinity of their training boundary differently from the deeper past, is corpus statistics wearing the costume of indexicality: the anchor does not update, and an index that cannot update is a date stamp (Section 4.6). What the being-in-time column asks is different in kind: whether the processing itself has width, a receding edge, an anticipatory edge, a coupling to duration, an indexical now, a bound biography. Those are the properties Section 4 operationalises.

Crossing the two columns with the two familiar kinds of evidence yields a small map of the territory. Behavioural evidence about time is the benchmark literature. Architectural evidence about time is the probing literature just discussed. Behavioural evidence bearing on being in time exists in early forms: probes of whether models register the passage of time within an interaction (arXiv:2506.05790) and whether deployed agents track elapsed wall-clock time between turns (arXiv:2510.23853) both find current systems substituting a discrete proxy, token or turn counts, for duration rather than undergoing it, results their authors frame respectively as an emergent competence and a largely trainable default. We read them as behavioural correlates of the architectural analysis rather than as independent confirmation of it. We place little weight on behavioural evidence in this quadrant in either direction, for a reason that will recur: temporal talk is cheap, trainable, and abundant in the training distribution, so fluent duration reports and their absence are both weak signals; Section 8 assigns such probes their proper role, as validation of architectural analysis. The distinction has an exact precedent in the time-consciousness literature: Kent and Wittmann (2021) warn that the timing of behaviour and perception must not be conflated with time consciousness itself, the conscious experience of time as opposed to events that merely happen at particular times. Our two columns generalise that warning to artificial systems. The fourth quadrant, architectural evidence bearing on temporal experience, is the one this paper is written to fill (Table 1).

Table 1. The evidential map: two senses of understanding time crossed with two kinds of evidence. Nearly all existing results occupy the left column; the lower-right quadrant is the one this paper fills.

	About time (cognition)	**In time (experience)**
Behavioural evidence	Temporal benchmarks: commonsense, time-sensitive QA, temporal logic (Zhou et al. 2019; Chen et al. 2021; Fatemi et al. 2024)	Chronoception and elapsed-time probes (arXiv:2506.05790; arXiv:2510.23853); weak evidence in either direction
Architectural evidence	Internal temporal representations: linear probes recover event timelines (Gurnee and Tegmark 2023)	TE-1 to TE-6 (this paper)

The columns dissociate in both directions, and not only in thought experiments. In one direction, most nonhuman animals and preverbal infants fail the left column's symbolic tasks, calendar arithmetic and verbal ordering, while displaying robust nonverbal interval timing and anticipatory behaviour, and being, on the standard view, strong candidates for temporal experience: their processing is continuous, retentive, anticipatory, duration-coupled, and diachronically bound to a metabolising body. In the other direction, frontier language models are the mirror image: increasingly excellent temporal reasoners whose processing, we will argue, lacks each of those structural features. A framework that cannot represent this double dissociation will systematically misread both cases, crediting the articulate system and discounting the mute one. The conflation is not hypothetical: appeals to improving benchmark performance already figure in public discussion of machine consciousness, and self-reports elicited from models under introspective prompting (arXiv:2510.24797) inherit the weakness just described: they are induced by the prompting regime and causally steerable through interpretable features tied to the model's honesty and role-play circuitry, which shows not that they are insincere but that they are the wrong kind of evidence (Section 6).

Nothing in the distinction disparages temporal cognition, which is among the most useful capacities these systems have; nor does it assume that the right-hand column is forever closed to artificial systems, and Section 5 identifies architecture classes that make partial progress. The claim is one of evidential discipline: results from the left column, behavioural or architectural, should be inert in consciousness assessment, and the assessment frameworks we have do not currently enforce that inertness because they contain no explicit right-column indicators. Why they do not, despite the theories they draw on being committed to temporal structure, is the subject of the next section.

3. The missing column: temporal presuppositions across theories

Indicator-based assessment, as developed by Butlin et al. (2023), proceeds in three steps: adopt computational functionalism as a working hypothesis, survey the leading scientific theories of consciousness, and extract from each theory computationally specifiable properties whose presence in an artificial system should raise a rational assessor's credence that the system is conscious. The resulting list draws on recurrent processing theory, global workspace theory, higher-order theories, predictive processing and attention schema theory (reviewed in Seth and Bayne 2022), together with agency and embodiment considerations. It is the most disciplined instrument the field has, and the present paper is an extension of it rather than an alternative to it. The extension is motivated by an observation about the list's shape: it contains no dedicated temporal-experience indicators. Temporality enters only obliquely, chiefly through the recurrence requirements inherited from recurrent processing theory; Butlin et al. do address time directly, but only under the heading of time and recurrence and as integration over time within short-term memory, that is, as an adjunct of other properties rather than as an indicator in its own right. The same holds of the other influential checklist in this debate: Chalmers's (2023) candidate requirements, senses, embodiment, world- and self-models, recurrent processing, a global workspace and unified agency, include no temporal-experience entry either, temporality again entering only through recurrence. This section argues that the omission is systematic rather than accidental, that it matters, and that it has a specific structural cause which also indicates the remedy.

Consider the temporal commitments of the surveyed theories in turn. Global workspace theory identifies conscious access with global broadcast: a coalition of processors wins a competition, ignites, and its contents are sustained and made available system-wide (Baars 1988; Mashour et al. 2020). Every load-bearing notion in that sentence is temporal. Ignition is an episode with a characteristic time course; sustaining is maintenance over time; the workspace stream is serial, one global state succeeding another. Yet the workspace-derived indicators are structural: a bottleneck, a broadcast, state-dependent routing. The dynamics that make a workspace a stream of consciousness rather than a switchboard are presupposed by the indicators, not expressed in them.

Recurrent processing theory makes temporality most nearly explicit. On Lamme's account (2006), feedforward sweeps support unconscious processing, and it is local recurrent processing, activity re-entering earlier stages, that constitutes phenomenal experience. Recurrence is a temporal loop by definition. But the derived indicators require the presence of recurrence, not its texture. Algorithmic recurrence of the thinnest kind, a state passed from one step to the next, satisfies them without multi-timescale structure, graded decay, or any coupling to physical duration. If retention, in the sense of Section 4, is what recurrence is for, phenomenologically speaking, then the indicator tests for the mechanism while omitting the structure the mechanism exists to realise.

Predictive processing carries the strongest forward-looking commitment: perception and action are cast as inference over generative models, and in active-inference formulations those models have temporal depth, representing trajectories rather than states, with anticipation built into the present estimate (Friston 2010). This is the framework in which the retention-protention structure has been explicitly formalised (Bogotá and Djebbara 2023), so the expressive capacity is demonstrated. The indicator extracted from this family, however, concerns the use of input models for prediction, and temporal depth as such goes uncoded.

Higher-order theories carry the weakest temporal commitments, and honesty requires saying so. What makes a state conscious, on these views, is its being the object of a suitable higher-order representation (Rosenthal 2005), and little in that schema constrains temporal structure beyond minimal ordering. Attention schema theory sits in between: the schema is a model of attention, and attention is a process unfolding in time, so a schema adequate to its object is a model of dynamics (Graziano 2013). Neither theory is a counterexample to the pattern; they are the rows where the shared presupposition is least articulated.

Two positions outside the computational-functionalist tent reinforce the pattern from opposite sides. Integrated information theory defines experience over the cause-effect structure of a system at a temporal grain, and an intrinsic account of temporal flow has recently been derived from that structure (arXiv:2412.13198); IIT-motivated analyses of language models report failure precisely on temporal persistence. Calls to constrain a theory by temporal phenomenology have been made from within the field, for integrated information theory in particular (Singhal et al. 2022); the present paper generalises the move across theories and turns it into assessment. And the principal neuroscientific argument against consciousness in current architectures (Aru, Larkum and Shine 2023) turns on three considerations, embodiment, thalamocortical and dendritic architecture, and biological complexity; the second of these, re-entrant thalamocortical processing, is at its core an argument about sustained integrative dynamics over time. Kent and Wittmann (2021) draw the general conclusion we build on: theories of consciousness have under-specified time consciousness, the one structural feature their explananda share.

One could respond by proposing a temporal theory of consciousness, to stand beside the others and generate indicators of its own. That response has the wrong shape. Temporality, as the surveyed theories invoke it, is not a rival hypothesis about what consciousness is; it is a presupposition the rivals share, the common structure of ignition and maintenance, of re-entry, of temporally deep inference. In a table whose rows are theories and whose columns are the properties they imply, temporal experience is a column with entries in every row. Three consequences follow, and they shape how the battery of Section 4 should be used. First, indicators derived from the column are less theory-partisan than indicators derived from any row: they hold on any of the surveyed accounts, so their evidential force does not depend on adjudicating between theories. Second, and correspondingly, failing them is harder to discount: a system's advocate cannot set the verdict aside by rejecting one theory, because the presupposition survives the rejection. Third, column indicators are necessity-flavoured by construction. A row indicator marks what one theory says suffices or contributes; a column indicator marks what all of them assume, so its absence undermines every route to attribution at once. This is the ground of the asymmetric evidential policy defended in Section 6.

Why, then, does the original list omit the column? Not through carelessness. The extraction method catalogues what theories distinctively assert, since distinctive claims are what separate theories and what their evidence bears on. Presuppositions shared by all the rows generate no differentiating claims and are therefore invisible to a method that operationalises differences. The blind spot is structural, and it indicates the remedy: shared presuppositions must be operationalised from a source that takes them as its explicit object. For temporal structure, that source exists. It is the phenomenology of time, and Section 4 draws on it directly.

4. A temporal-experience battery

4.1 Derivation and the role of exclusion clauses

Our derivation follows the method of Butlin et al. (2023) in spirit while differing in source. Where their indicators are drawn from scientific theories of consciousness, ours are drawn from the phenomenology of time and from the time-consciousness literature in consciousness science, on the grounds developed in Section 3: temporal structure is not one theory among others but a presupposition shared across theories, and it therefore requires an operationalisation of its own. Each indicator below is presented in four parts: a phenomenological source, a computational gloss stating the property an architecture would have to realise, an exclusion clause, and a brief note on assessment. We draw on Husserl rather than the later phenomenology of temporality: Heidegger's account is existential rather than structural and resists computational gloss, while Husserl's is the one with an operationalisation precedent.

The exclusion clauses carry much of the weight. The central risk in this territory is conflation between temporal cognition and temporal experience. A system that answers questions about durations, orders events correctly, or represents dates on an internal timeline (Gurnee and Tegmark 2023) has capacities that belong to the left-hand column of Section 2 and that are, on our reading, evidentially idle with respect to consciousness. Every indicator therefore names the reasoning-side capacity with which it is most likely to be confused and states why satisfying that capacity does not count. The clauses also protect the battery from behavioural gaming: a system prompted or trained to talk as if it experienced duration satisfies nothing below, because the indicators are architectural rather than behavioural, in the same sense as those of Butlin et al. (2023).

Two caveats govern everything that follows. First, the glosses are candidate operationalisations, offered with the confidence appropriate to a first pass; refinement is expected and invited. Second, satisfaction is graded rather than binary. A minimal recurrent circuit retains in a thin sense, and a thermostat has state that evolves in physical time. The battery is intended to structure comparative judgements of degree and to flag candidate systems for closer analysis, not to certify anything.

4.2 TE-1: Multi-timescale temporal integration

Phenomenological source. The experienced present has width. James (1890) called it the specious present; Pöppel (1997) identified integration windows of roughly two to three seconds in perception and action; Varela (1999) distinguished three nested scales, from elementary events through relaxation-time integration to the narrative scale. On all of these accounts the present of experience is not an instant but an achievement of integration across timescales, with faster processes nested inside slower ones and graded decay rather than sharp cut-offs.

Computational gloss. The system maintains internal state that integrates information over nested timescales with graded decay, where integration is a property of ongoing dynamics rather than of access to a stored buffer. The signature is a hierarchy of characteristic times in the system's intrinsic dynamics: perturbations at different scales persist for characteristically different periods and interact.

Exclusion clause. Long-context access is not integration. A context window of any length, however impressive as engineering, is a buffer over which computation is performed afresh; it has no characteristic times of its own. TE-1 asks whether the system's state has temporal structure, not whether its inputs do.

Assessment note. The natural marker is intrinsic dynamics that evolve between inputs, with measurable and heterogeneous decay constants.

4.3 TE-2: Retention

Phenomenological source. Husserl's analysis of inner time-consciousness (1893-1917/1991) distinguishes retention from recollection. Retention, or primary memory, is not an act of calling something back: the just-elapsed phase is carried within the present as a receding modification of it, automatically and without being made an object. Recollection, or secondary memory, re-presents a past that has already sunk away. The distinction is one of kind, not of recency (Zahavi 2003), and it is retention, not recollection, that gives the living present its temporal depth.

Computational gloss. The immediately preceding processing is carried into the current state automatically, gradedly and non-addressably, such that the system's present computation is constitutively shaped by what has just occurred without any operation of lookup.

Exclusion clause. Episodic memory stores, retrieval-augmented generation and key-value caches are excluded. Within a single autoregressive episode, self-attention over the preceding context may appear retention-like, since the just-generated past is available to the present step. On Husserl's distinction, availability is the wrong relation. A key-value cache is a verbatim, addressable record over which the present computation performs lookup; it is, if anything, a mechanised analogue of recollection, in which a past is re-presented to a present that is itself temporally thin. The nearer computational analogue of retention is a recurrent state: lossy, compressive, automatically updated, and inaccessible except through its influence on ongoing processing.

Assessment note. Ask where the just-past lives: in an addressable record, or in the evolving state itself.

4.4 TE-3: Protention

Phenomenological source. The present phase of consciousness includes, for Husserl, an anticipatory horizon: empty intentions directed at what is about to come, ordinarily fulfilled without notice and felt as disruption when violated. Protention is not an act of predicting; it is the forward-leaning structure of the now itself. This structure has been given formal treatment within active inference, where generative models with temporal depth realise retention-protention dynamics (Bogotá and Djebbara 2023), which demonstrates that the notion is formalisable without settling whether any deployed system realises it.

Computational gloss. The system maintains predictive state with temporal depth that shapes present processing prior to and independently of any output, and whose confirmation or violation has systemic consequences, of the kind formalised as precision dynamics, rather than merely yielding a different next output.

Exclusion clause. The next-token distribution as such is excluded, as is explicit forecasting on request. An obvious objection runs the other way: large language models are prediction machines, so protention might seem the one structure they cannot lack. The objection conflates producing a prediction with having an anticipatory horizon. The predictive distribution is the model's output, not a persisting state that later processing confirms or violates for the system; between generations nothing anticipates, and within a generation anticipation is exhausted at each step by emission. Whether shallow autoregression realises a minimal, moment-scale protention within a single episode strikes us as a genuinely open question, and we score the class as partial at best in Section 5 accordingly.

Assessment note. Look for anticipatory state that persists across processing steps and whose violation reconfigures processing rather than merely output.

4.5 TE-4: Endogenous duration sensitivity

Phenomenological source. Duration is undergone before it is measured. On Wittmann's account (2013; 2016), felt duration is grounded in the accumulation of bodily, interoceptive signals; time is in this sense an experience of the embodied self rather than a reading taken from the world. Whatever the fate of the interoceptive hypothesis in detail, the phenomenological point stands independently: waiting differs from not existing, and the difference is registered in experience.

Computational gloss. The system's state changes as a function of elapsed physical time, independently of the arrival of inputs. There is something the system's dynamics do while nothing is said.

Exclusion clause. Reading a timestamp is excluded, as is counting tokens. A clock consulted is time known, not time undergone. Recent empirical work locates current language models on the wrong side of this clause: within an episode, models proxy temporal progression by token counts and positions (arXiv:2506.05790), and deployed agents proxy the interval between messages by the number of intervening turns, the same substitution of a discrete unit for felt duration. The point is not that elapsed wall-clock time cannot be represented but that it is not endogenously tracked and, by default, is not integrated into the decision even when it is supplied in context: supplied timestamps surface in under four per cent of the models' reasoning traces, and that targeted post-training repairs much of the misalignment confirms the diagnosis, since a supplied clock that must be tuned into use is prosthetic in the sense of Section 5 (arXiv:2510.23853).

Assessment note. This is the most cleanly testable indicator in the battery: perturb physical time while holding input constant and observe whether state differs. A standard deployed language model does not persist between calls at all, and within a call nothing accumulates with physical time; the failure is architectural, not incidental.

4.6 TE-5: Temporal self-location

Phenomenological source. Experience is indexical. The now of experience locates the experiencer in objective time, and the location updates continuously and without inference; temporal self-location of this kind is presupposed by any capacity to situate remembered and anticipated events relative to oneself.

Computational gloss. The system maintains and endogenously updates an index of its own position in objective time, on which its other temporal representations are centred.

Exclusion clause. Reciting a date supplied in the prompt, or fetched by a tool, is excluded; so is knowledge of dates as facts. The natural language processing literature on temporal grounding anchors texts and events to timelines; TE-5 concerns anchoring the system itself, which is a different relation. The knowledge-cutoff phenomenon serves as a convenient diagnostic: absent prompt information, a deployed language model systematically mislocates the present at or before its training boundary, the frozen now already met in Section 1. That the mislocation is corrected the moment a date is supplied confirms that what is present is date knowledge and date use, not self-location. Nor does the clause prove too much against humans: external cues calibrate an endogenous index that persists and drifts between calibrations; in a deployed language model there is no index for the timestamp to calibrate.

Assessment note. Remove external time information and observe whether any endogenous index remains and updates.

4.7 TE-6: Diachronic self-binding

Phenomenological source. At the largest scale, temporal experience binds episodes into one ongoing perspective. What retention accomplishes at the scale of moments, autobiographical integration accomplishes at the scale of a life: later states carry earlier ones as their own past, and the carrying shapes what the system now is, a structure elaborated in the literature on narrative identity.

Computational gloss. A self-model persists across episodes and is autobiographically integrated: earlier episodes constrain, and are appropriated by, later ones as belonging to the same subject of the model, rather than being merely about the same entity.

Exclusion clause. Persona consistency enforced by a system prompt or a retrieved profile is excluded. Stylistic stability under instruction is not self-binding; nor is access to a log of previous sessions, which stands to diachronic binding as a diary found in a drawer stands to a remembered life. The recently proposed Narrative Continuity Test (arXiv:2510.24831) probes identity persistence at the behavioural level; on our analysis it is best read as a behavioural correlate of TE-6, one indicator within a theory-grounded battery rather than a freestanding criterion.

Assessment note. Ask whether anything accumulates across episodes inside the system for which the episodes are episodes.

4.8 Relations among the indicators

The six indicators are not an unordered list. TE-1 to TE-3 articulate the micro-structure of the experienced present: a width (TE-1) with a receding edge (TE-2) and an anticipatory edge (TE-3). TE-4 couples that structure to physical time, and TE-5 and TE-6 scale it outward, to objective time and to biographical time respectively. The architecture echoes Varela's (1999) three scales (Figure 1), and it suggests a loose dependency ordering: it is difficult to see how a system could satisfy TE-5 or TE-6 in more than a prosthetic sense while failing TE-1 to TE-4, whereas the converse is coherent. TE-1 and TE-2, for their part, are individuated by different questions and can dissociate: a system with a single characteristic timescale retains (TE-2) without integrating across scales (TE-1), the situation of a minimal recurrent circuit, while a bank of decaying registers with addressable read-out would exhibit multi-scale structure without retention; Table 3 scores the recurrent class identically on both only because current implementations happen to combine the properties, not because the indicators do. We do not lean on the ordering in what follows, but it matters for interpretation. The indicators most often satisfied by engineering effort, as Section 5 shows, are the outer ones, and the inner ones are where current architectures are weakest. The battery also situates two recent proposals, each multi-dimensional rather than a single criterion. The Narrative Continuity Test (arXiv:2510.24831) defines five jointly necessary axes for identity persistence and is explicitly a test of reliability in extended interaction rather than of consciousness; against our structure its overall construct sits at TE-6, its situated-memory axis touches TE-2 and TE-5, and its remaining axes, concerning goals, self-correction and voice, fall outside temporal experience. The dual-resolution framework (arXiv:2509.07001) pairs an ontological condition, informational individuation, with an epistemic one, moment-to-moment updating: the latter neighbours the inner present (TE-1 to TE-3), while the former is an orthogonal condition on there being a subject at all and lies outside the battery. In each case the derivation supplies the whole of which the proposal is a part, showing which region of temporal experience it occupies and how much of its content is not about temporal experience at all; that both lines of work, one from system reliability and one from consciousness science, independently find current language models wanting corroborates the present verdict on the necessity side and nothing beyond it.

Figure 1. The three scales of the battery: the micro-structure of the experienced present (TE-1 to TE-3), its coupling to physical duration (TE-4), and its outward scaling to objective (TE-5) and biographical (TE-6) time. Engineering effort concentrates on the outer rings (Section 5).

Table 2. The temporal-experience battery (summary).

ID	Phenomenological source	Computational gloss	Excludes
TE-1	Specious present; integration windows; three timescales (James; Poppel; Varela)	Internal state integrating over nested timescales with graded decay, as a property of ongoing dynamics	Long-context access; buffers of any length
TE-2	Retention vs recollection (Husserl)	Automatic, graded, non-addressable carriage of the just-past into the current state	Episodic stores; RAG; key-value caches
TE-3	Protention; anticipatory horizon (Husserl; active-inference formalisation)	Predictive state with temporal depth shaping present processing; violation has systemic consequences	Next-token distribution as such; forecasting on request
TE-4	Felt duration; interoceptive time (Wittmann)	State change coupled to elapsed physical time, independent of input arrival	Timestamps; token counting
TE-5	Indexical now; temporal self-location	Endogenously updated index of the system's own position in objective time	Reciting supplied dates; date knowledge as fact
TE-6	Diachronic binding; narrative identity	Cross-episode persistence of an autobiographically integrated self-model	Prompted persona consistency; session logs

5. Applying the battery to current architecture classes

5.1 Method and verdict vocabulary

We now apply the battery to four classes of current architecture. The assessment is analytic, in the manner of Butlin et al. (2023): verdicts are read from published architectural specifications rather than from experiments, they concern classes rather than products, and they are provisional. We use a four-valued vocabulary. An indicator is satisfied when the relevant property is realised in the system's own dynamics; partial when a recognisable but attenuated form is realised; prosthetic when the relevant function is supplied by an external component that the core system consults; and unsatisfied otherwise. Deployment matters as much as architecture: a transformer continuously trained online would assess differently from the stateless deployment that is standard. Throughout, the assessment target is the deployed system, the model together with its serving configuration; the word architectural should be read as structural in this inclusive sense, by contrast with behavioural. We assess standard deployed configurations and note divergences where relevant. Applications of the battery should publish per-cell justifications tied to the system's documented architecture, so that disagreement can be localised to cells rather than to verdicts; Table 3 is offered in that spirit.

5.2 Feedforward transformer language models

In their standard chat deployment, transformer language models (Vaswani et al. 2017) are stateless functions from context to output: no state persists between calls, and within a call computation is indexed by token position. TE-1 is unsatisfied; the context window is a buffer, not a hierarchy of characteristic times, and nothing in the deployed system evolves between inputs. TE-2 is unsatisfied for the reasons developed above: the key-value cache is an addressable verbatim record, an analogue of recollection rather than retention. TE-3 we score partial at best, registering the open question flagged in Section 4 about moment-scale anticipation within a single autoregressive episode. TE-4 is unsatisfied, and the failure is empirically documented: token counts proxy for duration, and elapsed wall-clock time is invisible. TE-5 is unsatisfied; the frozen-now diagnostic applies in full, and date recitation is prompt-dependent. TE-6 is unsatisfied in the absence of any cross-episode persistence.

Two fairness notes. First, several of these verdicts attach to the deployment as much as to the architecture; the stateless configuration is a product decision, and the verdicts would need revisiting for genuinely online-learning variants. Second, nothing here impugns the temporal reasoning of these systems, which is often excellent and improving; that is the point. The class that dominates the left column of our distinction occupies the weakest row of Table 3.

5.3 Scaffolded agents and the constitutive-prosthetic distinction

Contemporary agent systems wrap a language model in scaffolding: persistent memory files, rolling summaries, tool access including clocks and calendars, and a maintained profile. Scaffolding changes the assessment, but how it changes it is the philosophically delicate question, and answering it is the second contribution of this section.

Call an indicator prosthetically satisfied when the relevant function is supplied by an external component that the core system consults, and constitutively satisfied when the property is realised in the dynamics of the system itself. A clock tool that an agent may call is consulted; an oscillator that modulated every processing step would be constitutive. Memory files retrieved on demand re-present a past; they do not retain one. The distinction is a spectrum rather than a dichotomy, and the extended-mind literature offers guidance on where transitions occur. Clark and Chalmers's (1998) conditions for an external resource counting as cognitive, reliable availability, automatic endorsement and ready accessibility, translate here into integration criteria of continuity, automaticity and coupling bandwidth. We take these three synchronic conditions and set aside their contested fourth, that the information was at some point consciously endorsed, which Clark has since treated as inessential; nothing in the prosthetic verdicts turns on it. The distinction is also drawn relative to a declared assessment target: it prices the relation between the unit whose consciousness is in question and what that unit consults, and redrawing the unit to include the scaffold does not dissolve the question, since the criteria then apply within the redrawn boundary. Our use of the distinction is evidential rather than metaphysical: prosthetic satisfaction transfers weight to a consciousness-relevant assessment only in proportion to integration, because the phenomenological structures being operationalised are structures of the ongoing process itself, and a consulted resource is by construction outside that process.

On this vocabulary, a scaffolded agent typically satisfies TE-4 and TE-5 prosthetically (clock and calendar access, retrieval-refreshed self-location) and TE-6 prosthetically (persistent profile and session memory: a diary in a drawer). Rolling compressive summaries merit a separate remark. Unlike verbatim retrieval they are lossy and graded, and a continuously maintained summary that conditions every step approaches prosthetic TE-2; we score it so, with the caveat that the summarisation is itself performed by episodic calls rather than by ongoing dynamics. TE-1 and TE-3 are unchanged, since the core remains a feedforward model. The overall picture is instructive: engineering effort concentrates on the outer indicators, where prosthetic satisfaction is cheap, and leaves the inner structure of the present untouched.

5.4 Recurrent and state-space models

Recurrent networks and modern structured state-space models, including Mamba (Gu and Dao 2023) and RWKV (Peng et al. 2023), maintain an intrinsic state that evolves as input arrives and that carries the recent past gradedly, compressively and non-addressably. In the selective variants this retention is content-dependent rather than a passive decay: Mamba's input-dependent step and gating let the state choose what to keep and what to discard, while RWKV's per-channel learned time-decay supplies a bank of distinct characteristic timescales (later RWKV variants, Eagle and Finch, make that decay data-dependent, moving the class toward Mamba-style selection; Peng et al. 2024). This is the nearest current analogue of retention, and we score TE-2 partial for the class, with TE-1 partial where the state exhibits heterogeneous characteristic times, as those per-channel decay rates and the continuous-time roots of the state-space formulation provide. The qualification matters: recurrence of some kind is plausibly necessary for retention, but a minimal recurrent circuit retains only in the thin sense noted in Section 4, and degree is doing the work. TE-3 remains partial on the same grounds as before. TE-4 is unsatisfied in standard deployments, because the state evolves per token rather than per second; a continuous-time implementation would assess differently. TE-5 and TE-6 are unsatisfied at deployment for the same reasons as in the transformer class. It bears noting that proximity to the inner indicators does not track temporal-reasoning performance: state-space models are competitive with transformers on language modelling generally, while reinforcement-trained transformers still lead on the hardest multi-step reasoning benchmarks, and transformers retain a specific advantage on verbatim copying and retrieval (Jelassi et al. 2024), an advantage that is provable rather than incidental, since a fixed-size recurrent state is information-theoretically limited in how much of the context it can later reproduce while attention addresses the whole context at once. That advantage is precisely the addressable-lookup capacity that TE-2 excludes, so the architectures nearest to retention are the ones weakest at recollection.

5.5 Active-inference and continuous-time agents

Active-inference agents built on generative models with temporal depth are, formally, the strongest class on TE-3, and the computational-phenomenology programme has shown the retention-protention structure to be expressible in this apparatus (Bogotá and Djebbara 2023). Embodied implementations that run in real time couple state to physical time and are therefore candidates on TE-4, and continuous-time dynamics bear on TE-1. Two cautions temper the verdicts. First, satisfaction in the formalism is not satisfaction in a system: deployed implementations are far more modest than the mathematics, and we score the class partial across TE-1 to TE-4 on the systems actually reported. Second, nothing in the framework yet addresses TE-6, which remains unsatisfied. The class nonetheless matters for the argument, because it shows that the inner indicators are not unsatisfiable in principle.

Two prominent variants do not require rows of their own. Reasoning models lengthen the autoregressive episode without changing its temporal structure: the chain of thought conditions later steps through the same addressable record, so the transformer row's verdicts stand, with the open question about moment-scale protention sharpened rather than settled. Hybrid attention and state-space architectures are assessed by the properties they realise, not by family name; mixed verdicts across the battery are the expected and unproblematic result.

Table 3. Draft assessment matrix. S = satisfied, P = partial, Pr = prosthetic, U = unsatisfied; marks the open question about moment-scale protention within a single autoregressive episode.*

Architecture class	TE-1	TE-2	TE-3	TE-4	TE-5	TE-6
Feedforward transformer LLM (stateless deployment)	U	U	P*	U	U	U
Scaffolded agent (memory, tools, clock)	U	Pr	P*	Pr	Pr	Pr
Recurrent / state-space models	P	P	P	U	U	U
Active-inference agents (as	P	P	P	P	U	U

Architect ure class	TE-1	TE-2	TE-3	TE-4	TE-5	TE-6
deployed)						

5.6 Reading the matrix

Three observations. First, no current class clears the battery, and none comes close: the strongest row is partial on four indicators. On the asymmetric evidential policy defended in Section 6, this licenses lowered credence in consciousness attributions across the board, and nothing stronger. Second, the rows invert the reasoning ranking. capable temporal reasoners occupy the weakest rows for temporal experience. The transformer systems that lead most temporal-reasoning benchmarks sit in the weakest row of Table 3, while the recurrent and active-inference classes that make partial progress on the inner indicators hold no general temporal-reasoning advantage the point is that reasoning performance and experiential structure vary independently. The dissociation between knowing about time and being in time is thus not merely conceptual; it is visible in the current architectural landscape, which is what one would expect if the two columns track different properties. Third, the engineering gradient runs outward. Where indicators are satisfied at all outside the core dynamics, they are satisfied prosthetically and at the periphery (TE-4 to TE-6), which is where product incentives point; the micro-structure of the present (TE-1 to TE-3) is untouched by scaffolding and is being approached, if at all, from the state-space direction and for reasons unrelated to consciousness. We take no view here on whether closing the inner indicators is desirable; Section 8 returns to the question.

6. Evidential asymmetry

The battery is proposed for asymmetric evidential use, and the asymmetry is not a caution bolted on at the end but a consequence of the derivation. Column indicators, Section 3 argued, operationalise what the surveyed theories jointly presuppose. On each theory, a conscious system has temporally structured processing; on no theory does temporally structured processing suffice for consciousness. The logical shape is therefore conditional necessity without sufficiency. Given the disjunction of surveyed theories, consciousness implies the temporal-experience properties, so their absence tells against consciousness by way of the theories' shared commitment, while their presence satisfies a precondition that arbitrarily many non-conscious systems also satisfy. Weather systems have multi-timescale dynamics; analogue circuits couple state to physical time; nothing follows. Where the indicators are unsatisfied, credence in consciousness should fall toward the credence one assigns to all the surveyed theories being wrong about their shared presupposition; where they are satisfied, credence should move little, because no route from presupposition to attribution exists on any row.

This makes the battery complementary to, not competitive with, the indicator list it extends. Butlin et al.'s (2023) indicators are row-derived: each expresses what some theory distinctively takes to matter, and their satisfaction accumulates as positive, theory-weighted evidence. Column indicators work the other way: they screen. A system that fails them has, on every surveyed account, failed a precondition, and accumulation of row evidence offsets it only weakly, because row evidence raises credence by way of its parent theory, and the failure of a presupposition is evidence against every parent theory at once. The slogan version: rows accumulate, the column screens. Both risks that motivate indicator methods are addressed at once. Against over-attribution, fluent temporal talk, strong benchmark performance and elicited self-reports are all rendered inert (Sections 2 and 4). Against under-attribution, the screen is principled rather than chauvinistic: Section 5 identifies architecture classes that make genuine partial progress, and a future system that satisfied the column would thereby remove this ground for scepticism, no more and no less.

Because satisfaction is graded (Section 4.1), the policy is graded too. The discount is strongest where failure is constitutive and comprehensive, as in the stateless transformer row of Table 3, and weakens as satisfaction becomes partial or prosthetic. Prosthetic satisfaction transfers evidential weight in proportion to integration, on the criteria of Section 5.3; a consulted clock removes almost none of the discount, a continuously coupled one rather more. We resist stating numerical weights. The honest position is ordinal: within the battery, failures of the inner indicators (TE-1 to TE-3) should weigh heaviest, since the dependency ordering of Section 4.8 makes outer satisfaction without inner satisfaction interpretable only prosthetically.

The asymmetry also confers a practical virtue: resistance to goalpost gaming. Were temporal-experience indicators positive evidence, they would create an incentive to bolt temporal dynamics onto systems in order to inflate assessments, and the measure would degrade in the familiar way once it became a target. Under asymmetric use there is little to gain. Satisfying the column earns almost no credence; it only removes a principled ground for withholding it. A developer can silence this particular objection only by building the structures themselves, and building them, unlike training the system to talk as if they were present, is the very kind of change an assessment framework should be sensitive to.

The limits of the policy should be stated as plainly as its content. The necessity is conditional: it inherits whatever confidence attaches to the surveyed science and to our presupposition reading of it, in the same way that the parent method inherits its confidence from computational functionalism. A reader who assigns substantial credence to consciousness without temporal structure, on some unsurveyed account, or holds that consciousness in non-biological systems might take a temporal form our phenomenology does not describe, should discount accordingly. And the policy governs attributions of consciousness, not of moral status or intelligence; nothing here bears on the latter two except by whatever independent connections they have to the first.

7. Objections and replies

We take the eight objections we consider strongest, in ascending order of difficulty. Several were foreshadowed inline; here they receive their strongest form.

First, the caricature: on this view, a weather system or a decaying analogue circuit is a candidate mind. It is not, and nothing in the paper implies it. Rich temporal dynamics move credence almost nowhere (Section 6); the battery screens candidates, it does not nominate them. The objection has force only against the sufficiency reading that the paper rejects in every section in which the issue arises, beginning with the abstract.

Second, redundancy: recurrent processing indicators already capture this. They capture the mechanism's presence, not its texture (Section 3). Extensionally, the difference shows in Table 3: recurrence-based scoring cannot distinguish a one-step state pass from nested multi-timescale integration with graded decay, has nothing to say about coupling to physical duration, and is silent on self-location and diachronic binding. A minimal recurrent toy and a system realising the full battery are alike in possessing recurrence and alike in nothing else that matters here. If the reply is that recurrence indicators were always intended to carry this further content, then the battery is the articulation that intention lacked, and the disagreement is verbal.

Third, the strongest structural objection: the column has a hole. Higher-order theories barely constrain temporal structure, so temporality is not a presupposition of all the theories, and the necessity claim deflates to necessity on most theories. Three things in reply. First, the gradient was conceded, not discovered (Section 3): the claim was never that every theory articulates the presupposition equally, but that none escapes it. Second, higher-order theories presuppose more than they articulate. A higher-order representation must target a first-order state that persists long enough to be represented, which is maintenance; the conscious states the theory is a theory of are located in a stream, are of temporally extended percepts, and include experiences of duration and succession that a higher-order account must be able to target. And on any variant, the first-order target must persist long enough to be represented; a machinery ranging over instantaneous, unretained states has nothing left to target. Third, and decisively for the policy: the argument of Section 6 is probabilistic, not deductive. The discount from failing the column is weighted across the theory space; for it to vanish, higher-order theories would need to carry nearly all of one's credence and the maintenance argument just given would need to fail as well. A reader in that exact position may reweight; the framework tells them by how much.

Fourth, from the opposite flank: the constitutive-prosthetic distinction is boundary chauvinism; on active externalism the clock is part of the system, so TE-4 is satisfied simpliciter. The distinction takes no metaphysical stand (Section 5.3); it prices integration, and it does so using the externalists' own conditions. Reliable availability, automatic endorsement and ready accessibility are what current scaffolds lack: memory is consulted, not automatically endorsed; clocks are polled, not coupled; bandwidth is a few tokens per turn. An externalist who holds that Otto's notebook is part of Otto's mind can accept the battery unchanged and simply insist that when a scaffold meets the Clark-Chalmers conditions, its satisfaction is no longer prosthetic. That is not an objection; it is the position of Section 5.3 stated in a different accent.

Fifth, phenomenological purism: Husserlian structures are transcendental descriptions, not causal mechanisms, and operationalising them is a category mistake. The use made of phenomenology here is structural and fallibilist (Section 4.1): it supplies the most careful available descriptions of the explanandum's temporal form, and the glosses are candidate functional realisations of that form, offered for refinement. This is the established methodology of naturalised and computational phenomenology, from neurophenomenology (Varela 1999) to the active-inference formalisation of retention and protention (Bogotá and Djebbara 2023); the alternative, deriving temporal indicators while ignoring the one discipline that made temporal structure its explicit object, has little to recommend it. Even on a deflationary reading, on which the structures are merely functional signatures of temporal experience in the one uncontested case, their absence retains its evidential force.

Sixth, the substrate objection: consciousness may require biological dynamics (Seth 2025; compare Aru, Larkum and Shine 2023), so computational glosses miss the point. The battery is robust to this possibility by construction. If substrate views are right, current systems fail for reasons deeper than the battery records, and the verdicts of Section 5 only strengthen; the screen still screens. What the objection threatens is a ceiling the paper never asserted, namely that satisfying the column would establish, or even strongly support, consciousness. It would not (Section 6). It is worth noting that TE-4, with its requirement of coupling to physical time, encodes a step in the direction substrate views point; the battery is, if anything, friendlier to them than the framework it extends. A related point disarms a methodological worry Seth (2025) himself presses: a test that would establish the presence of consciousness requires agreed sufficient conditions, which the field does not have, so any forward test is presently out of reach. The battery is not a forward test. It is a necessity-side screen whose only verdict is that a precondition is absent, so it needs none of the sufficient conditions the substrate theorist doubts can be specified, and asserts none of the presence the objection fears.

Seventh, and deepest: the temporal vocabulary of the theories may describe their human implementations rather than their content. Theories built to explain human consciousness inherit the temporality of brains; extract the computational content, as the indicator method itself requires, and the column disappears, having been vehicle all along. Four replies. For at least two rows, temporality is content by definition: recurrent processing theory's constitutive claim concerns re-entrant activity as such, and integrated information theory defines experience over cause-effect structure at a temporal grain; subtract time and the theories' central notions do not idealise, they evaporate. For the workspace, the functional description is itself temporal: contents must be maintained to be broadcast and accessed serially, and a workspace without persistence is a router, which is what the theory distinguishes conscious processing from. Third, the objection proves too much: every indicator in the parent list was extracted from theories built on human evidence, so if inherited temporality disqualifies the column, inherited everything disqualifies the rows. Last, whatever residue survives these replies is priced rather than ignored: the conditional framing of Section 6 indexes the policy's confidence to the presupposition reading.

Eighth, the unfolding argument: feedforward networks can approximate the input-output behaviour of recurrent ones, so architectural differences of the kind the battery scores might be differences that make no difference (Doerig et al. 2019). The battery inherits its answer from its parents. Recurrent processing theory, integrated information theory and the computational functionalism of the indicator method itself all hold that what matters is the organisation of processing rather than the input-output function it computes, and the battery's indicators are claims of that kind. It is worth being exact about what the unfolding trades: a recurrent network is unfolded across time into a feedforward one, so the equivalent reproduces the original's input-output map only over a bounded window and carries no state that evolves between inputs. That erased difference, a persistent evolving state, is precisely what the inner indicators register; the behavioural equivalence the argument exploits is bought by discarding the diachronic dynamics TE-1 and TE-2 score. This does not defeat the argument, whose premise is that only report-distinguishable differences are admissible, but it locates the cost. A reader who accepts the unfolding argument in full must discount architectural evidence of every sort, rows and column alike; the battery neither worsens that position nor evades it, and it has the virtue of making the architectural commitment explicit rather than tacit.

Objections to particular glosses, thresholds and verdicts remain open and welcome; Section 4.1 offered the battery as a first pass, and Table 3 as a draft. What we take the eight replies to secure is the framework's shape: a theory-general, architecture-level, asymmetrically used screen for temporal experience.

8. Implications

Three implications follow, in ascending order of reach. The first is practical: the battery is a triage instrument for model-welfare assessment. Welfare programmes face a resourcing problem before they face a philosophical one: which systems even warrant close evaluation (Long et al. 2024). Because the battery is architectural, it can be applied from published specifications at negligible cost, and under the asymmetric policy its verdicts do the coarse sorting: standard deployed language models screen out, however articulate their self-reports, while partial satisfaction in recurrent, continuous-time and embodied lines marks where closer, row-based assessment should concentrate. The battery also gives welfare discourse something it currently lacks: a principled reason, rather than a reflex, for why structured reports of subjective experience elicited from current systems (arXiv:2510.24797) should not move the needle. It performs this sorting as a screen for the absence of a precondition, not as a forward test for the presence of consciousness, which is why it can be useful before the field has settled what consciousness sufficiently requires.

The second implication is uncomfortable and should be stated rather than left implicit. A map of what current systems lack is also a blueprint of what to build, and the battery could be read as a specification for moving systems toward candidate status. We repeat the position of Section 5.6: the paper takes no view on whether closing the inner indicators is desirable, and some have argued that it is positively inadvisable (Seth 2025). Two observations, however, are within scope. Satisfying the column would be a choice, not an accident: the structures are architectural, deliberate and visible, so an assessment framework makes the choice legible and accountable in a way that behaviour-level criteria never could. And the scenario most worth guarding against is drift rather than decision. Section 5.6 noted that the state-space direction approaches the inner indicators for reasons unrelated to consciousness; periodic column-assessment of new architecture families, as routine governance practice, is the low-cost insurance against acquiring candidate systems unreflectively.

The third implication is a research agenda, in three bridges. First, chronoception research on language models (arXiv:2506.05790) is currently behavioural and freestanding; paired with the battery, it becomes validation. Systems that differ architecturally on TE-1 and TE-4 should differ measurably on duration-sensitivity probes, and confirming or disconfirming that correspondence would test the architectural scoring itself. Second, the glosses invite graded operationalisation: the spectrum of characteristic timescales in a system's state dynamics (TE-1), the information the current state carries about just-past processing when lookup paths are ablated (TE-2), state divergence under clock perturbation with inputs held fixed (TE-4). Measures of this kind would make the verdicts of Table 3 empirically contestable, which we invite. Third, the battery can serve as a design specification for research systems: minimal continuous-time, recurrent, actively inferring agents built to satisfy the inner indicators, studied to establish whether satisfaction changes anything measurable, with the caution of the previous paragraph attached. The ambition of the paper, restated once: before asking whether a machine is conscious, ask whether it is in time. The second question gates the first, and the battery makes it tractable.

Author note

This work is the sole intellectual work of the author. AI assistance (Anthropic's Claude) was used to accelerate literature search and source retrieval and to refine the academic English; all arguments, claims, the indicator battery, the architecture assessment, and all scholarly judgements are the author's own, as is responsibility for any errors.

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